

Test Report TR-2026-031 — KESTREL Prototype P2

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1. Summary

Six P2 prototypes were built with an rPP-50 seat shell, a recycled-aluminium base and zinc die-cast armrest brackets (part KS-204). Nine tests were run against the internal targets. Seven tests passed and two failed: armrest cyclic fatigue (T5) and assembly time (T9).

2. Results

ID	Test	Method	Target	Result	Status
T1	Seat drop impact	TP-02	No damage after 10 drops (75 kg)	No damage (3/3 units)	PASS
T2	Seat shell cyclic	TP-07	120,000 cycles	120,000 cycles, no failure (3/3 units)	PASS
T3	Backrest cyclic	TP-08	120,000 cycles	120,000 cycles, no failure (3/3 units)	PASS
T4	Armrest vertical static	TP-11	750 N, no failure	750 N, no failure (3/3 units)	PASS
T5	Armrest cyclic	TP-12	60,000 cycles	Crack at 38,500 cycles (unit 2) and 41,200 cycles (unit 4)	FAIL
T6	Base static load	TP-14	11,120 N, no failure	11,120 N, no failure (3/3 units)	PASS
T7	Rear stability	TP-16	No tip-over	No tip-over (3/3 units)	PASS
T8	Caster durability	TP-18	98,000 cycles	98,000 cycles, no failure (3/3 units)	PASS
T9	Assembly time	5 untrained testers	≤ 10 min	Mean 11.6 min (range 9.1–14.2 min)	FAIL

3. Failure analysis — T5 armrest cyclic

Unit 2 failed at 38,500 cycles and unit 4 at 41,200 cycles; the remaining unit reached 60,000 cycles without failure. In both failures a fatigue crack initiated at the R0.5 mm inner radius of bracket KS-204, next to the rear mounting boss.

Sectioning of unit 2 showed gas porosity up to 0.4 mm at the crack origin. A linear FEA model of the TP-12 load case gives a peak stress of 212 MPa at the radius.

Hypotheses: H1 — the sharp R0.5 mm radius concentrates stress; H2 — casting porosity reduces fatigue strength; H3 — loss of bolt preload lets the bracket rock. H3 is considered unlikely because the bolt torque was still 92% of nominal after the test.

4. Measurements

Measure	Value
Base diameter	592 mm
Seat width	475 mm
Mass (without packaging)	13.2 kg
Carton volume	0.118 m ³
Seat height range	420–530 mm

5. Assembly time (T9)

Five untrained testers assembled the chair from the carton with the included hex key. Times were 9.1 min, 10.8 min, 11.4 min, 12.5 min, 14.2 min. The mean of 11.6 minutes exceeds the 10-minute target. Mounting the backrest to the mechanism was the slowest step for all testers.

6. Recommendations

Increase the inner radius of the armrest bracket and eliminate porosity, or change the bracket process. Candidate fixes are option A (zinc, R2.0 radius plus rib), option B (stamped steel) and option C (flip-up armrest assembly). Costing is provided by Sourcing. Whichever option is chosen, T5 must be repeated on P3 with at least three units.

Pre-fit captive screws on the backrest to reduce assembly time, then repeat T9.